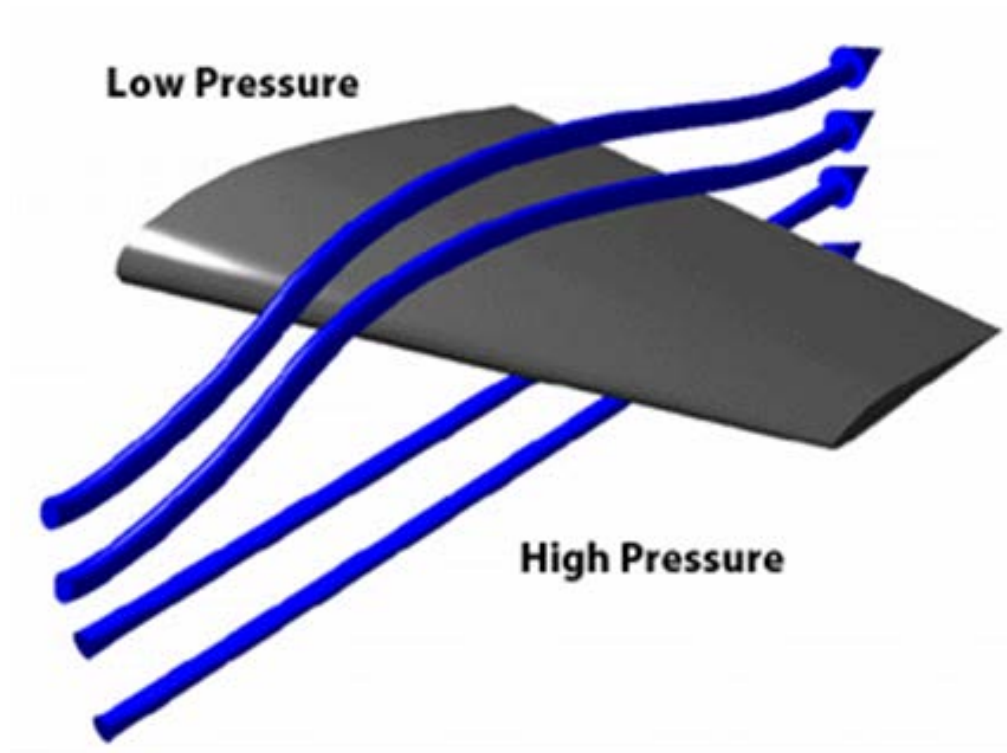


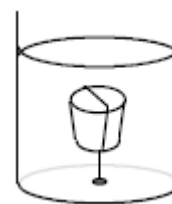
Chapter 15

Fluid Mechanics



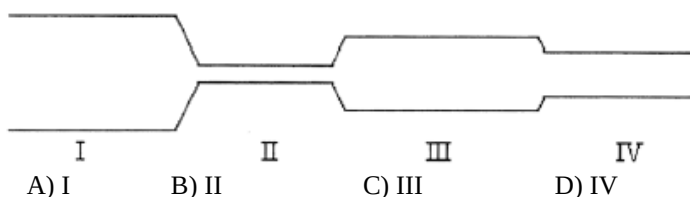
AP Physics Multiple Choice Practice – Fluid Mechanics

1. A cork has weight mg and density 25% of water's density. A string is tied around the cork and attached to the bottom of a water-filled container. The cork is totally immersed. Express the tension in the string in terms of the cork weight mg .
- A) 0
B) mg
C) $2mg$
D) $3mg$



2. An ideal fluid flows through a long horizontal circular pipe. In one region of the pipe, it has radius R . The pipe then widens to radius $2R$. What is the ratio of the fluid's speed in the region of radius R to the speed of the fluid in region with radius $2R$?
- A) 4 B) 2 C) $\frac{1}{2}$ D) $\frac{1}{4}$

3. A fluid is forced through a pipe of changing cross section as shown. In which section would the pressure of the fluid be a minimum?



4. Three objects all float on top of water. They have the following relationships:
- A and B have the same mass and same density but different shapes
 - B and C have the same volume and same shape
 - mass & density of C < mass & density of B
- Three identical weights are tied to each object, and each is pulled completely beneath the water. Which object will displace the greatest amount of water?
- A) A
B) B
C) C
D) All displace the same amount of water.
5. As a rock sinks deeper and deeper into water of constant density, what happens to the buoyant force on it?
- A) It increases.
B) It remains constant.
C) It decreases.
D) It may increase or decrease, depending on the shape of the rock.
6. A piece of wood with a volume of 50 cm^3 is floating on water, and a piece of iron with a volume of 50 cm^3 is totally submerged. Which has the greater buoyant force on it?
- A) The wood.
B) The iron.
C) Both have the same buoyant force.
D) Cannot be determined without knowing their densities.
7. Salt water is more dense than fresh water. A ship floats in both fresh water and salt water. Compared to the fresh water, the amount of water displaced by the ship when it is in the salt water is
- A) more.
B) less.
C) the same.
D) Cannot be determined from the information given.

8. Water flows through a horizontal pipe. The diameter of the pipe at point B is larger than at point A. Where is the water pressure greater?
- Point A
 - Point B
 - Same at both A and B
 - Cannot be determined from the information given.

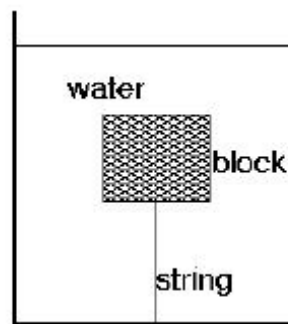
9. Liquid flows through a 4 cm diameter pipe at 1.0 m/s. There is a 2 cm diameter constriction in the line. What is the velocity in this constriction?
- 0.25 m/s
 - 0.50 m/s
 - 2 m/s
 - 4 m/s

10. A copper block is connected to a string and submerged in a container of water.
- Position 1: The copper is completely submerged, but just under the surface of the water.
- Position 2: The copper is completely submerged, mid-way between the water surface and the bottom of the container.
- Position 3: The copper is completely submerged, but just above the bottom surface of the container.

Assume that the water is incompressible. What is the ranking of the buoyant forces (B) acting on the copper blocks for these positions, from least to greatest?

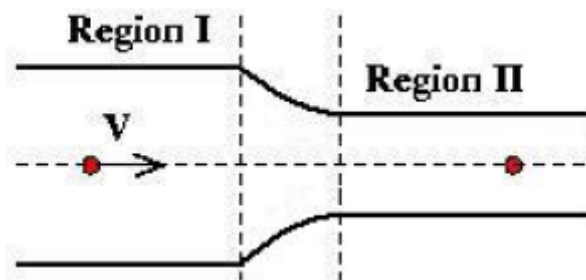
- $B_1 < B_2 < B_3$
 - $B_3 < B_2 < B_1$
 - $B_1 = B_2 = B_3$
 - $B_1 < B_2 = B_3$
11. Two objects labeled K and L have equal mass but densities $0.95D_o$ and D_o , respectively. Each of these objects floats after being thrown into a deep swimming pool. Which is true about the buoyant forces acting on these objects?
- The buoyant force is greater on Object K since it has a lower density and displaces more water.
 - The buoyant force is greater on Object K since it has lower density and lower density objects always float “higher” in the fluid.
 - The buoyant force is greater on Object L since it is denser than K and therefore “heavier.”
 - The buoyant forces are equal on the objects since they have equal mass.

12. A block is connected to a light string attached to the bottom of a large container of water. The tension in the string is 3.0 N. The gravitational force from the earth on the block is 5.0 N. What is the block’s volume?
- $2.0 \times 10^{-4} \text{ m}^3$
 - $3.0 \times 10^{-4} \text{ m}^3$
 - $5.0 \times 10^{-4} \text{ m}^3$
 - $8.0 \times 10^{-4} \text{ m}^3$



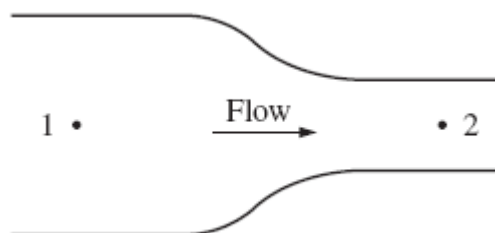
13. A cube of unknown material and uniform density floats in a container of water with 60% of its volume submerged. If this same cube were placed in a container of oil with density 800 kg/m^3 , what portion of the cube’s volume would be submerged while floating?
- 33%
 - 50%
 - 58%
 - 75%

14. The speed of an ideal fluid is marked as it moves along a horizontal streamline through a pipe, as shown in the figure. In Region I, the speed of the fluid on the streamline is V . The cylindrical, horizontal pipe narrows so that the radius of the pipe in Region II is half of what it was in Region I. What is the speed of the marked fluid when it is in Region II?



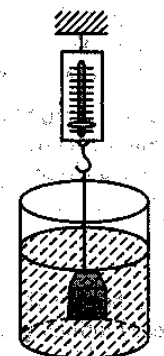
(A) $4V$ (B) $2V$ (C) $V/2$ (D) $V/4$

15. A fluid flows steadily from left to right in the pipe shown. The diameter of the pipe is less at point 2 than at point 1, and the fluid density is constant throughout the pipe. How do the velocity of flow and the pressure at points 1 and 2 compare?



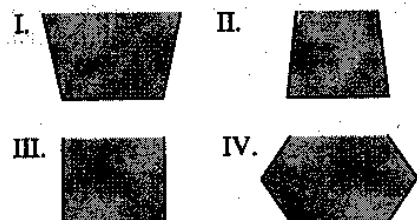
Velocity	Pressure
(A) $v_1 < v_2$	$p_1 = p_2$
(B) $v_1 < v_2$	$p_1 > p_2$
(C) $v_1 = v_2$	$p_1 < p_2$
(D) $v_1 > v_2$	$p_1 = p_2$

16. The figure shows an object of mass 0.4 kg that is suspended from a scale and submerged in a liquid. If the reading on the scale is 3 N , then the buoyant force that the fluid exerts on the object is most nearly



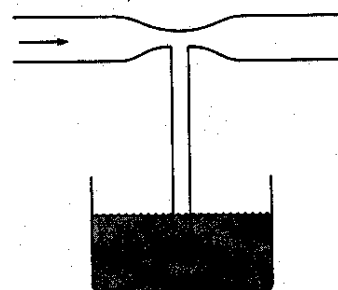
(A) 1.3 N
 (B) 1.0 N
 (C) 0.75 N
 (D) 0.33 N

17. Each of the beakers shown is filled to the same depth h with liquid of density ρ . The area A of the flat bottom is the same for each beaker. Which of the following ranks the beakers according to the net downward force exerted by the liquid on the flat bottom, from greatest to least force?



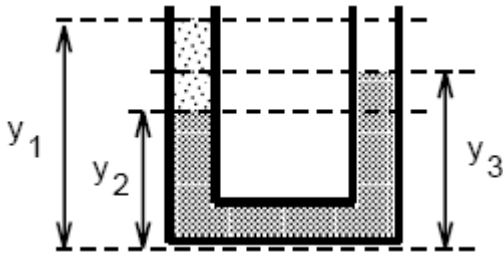
(A) I, III, II, IV
 (B) I, IV, III, II
 (C) II, III, IV, I
 (D) None of the above, the force on each is the same.

18. **Multiple Correct:** A T-shaped tube with a constriction is inserted in a vessel containing a liquid, as shown. What happens if air is blown through the tube from the left, as shown by the arrow in the diagram? Select two answers.



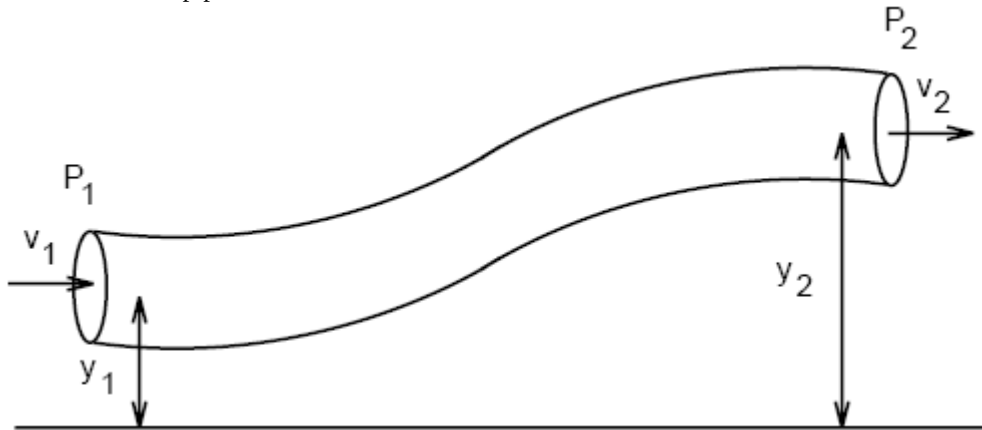
- (A) The liquid level in the tube rises to a level above the surface of the liquid in the surrounding tube
 (B) The liquid level in the tube falls below the level of the surrounding liquid
 (C) The pressure in the liquid in the constricted section increases.
 (D) The pressure in the liquid in the constricted section decreases.
19. One cubic centimeter of iron (density $\sim 7.8 \text{ g/cm}^3$) and 1 cubic centimeter of aluminum (density $\sim 2.7 \text{ g/cm}^3$) are dropped into a pool, and they sink to the bottom. Which has the larger buoyant force on it?
 (A) iron (B) aluminum (C) both are the same. (D) neither has a buoyant force on it
20. One kilogram of iron (density $\sim 7.8 \text{ g/cm}^3$) and 1 kilogram of aluminum (density $\sim 2.7 \text{ g/cm}^3$) are dropped into a pool, and they sink to the bottom. Which has the larger buoyant force on it?
 (A) iron (B) aluminum (C) both are the same. (D) neither has a buoyant force on it
21. Find the approximate minimum mass needed for a spherical ball with a 40 cm radius to sink in a liquid of density $1.4 \times 10^3 \text{ kg/m}^3$
 (A) 37.5 kg (B) 375 kg (C) 3750 kg (D) 37500 kg
22. A horizontal pipe of radius $7R$ carries a uniformly dense liquid to a spigot of radius R , where it has a speed of V . What is the speed of the liquid in the larger diameter pipe?
 (A) $0.02V$ (B) $0.11V$ (C) V (D) $49V$
23. The pressure in a pipe carrying a liquid with a density of ρ and an initial velocity v at the inlet is P , which is y meters lower than its outlet, which has a velocity of $2v$. In these terms, what is the final pressure?
 (A) $\frac{P}{2} \rho (3v^2 + 2gy)$
 (B) $P - \frac{1}{2} \rho (3v^2 + 2gy)$
 (C) $P + \frac{1}{2} \rho (-3v^2 + \rho gy)$
 (D) $\frac{\frac{1}{2} \rho (v^2 - 4v^2) - \rho gy}{P}$
24. A block of mass m , density ρ_B , and volume V is completely submerged in a liquid of density ρ_L . The density of the block is greater than the density of the liquid. The block
 (A) floats, because $\rho_B > \rho_L$
 (B) experiences a buoyant force equal to $\rho_B gV$.
 (C) experiences a buoyant force equal to $\rho_L gV$.
 (D) experiences a buoyant force equal to $m_B g$

25. **Multiple Correct:** In the open manometer shown, water occupies a part of the left arm, from a height of y_1 to a height of y_2 . The remainder of the left arm, the bottom of the tube, and the right arm to a height of y_3 are filled with mercury.



Which of the following is correct? (Select two answers.)

- (A) the pressure at a height y_3 is the same in both arms.
 - (B) the pressure at a height y_2 is the same in both arms.
 - (C) the pressure at the bottom of the right arm is the same as at the bottom of the left arm.
 - (D) the pressure at a height y_3 is less in the left arm than in the right arm.
26. Water flows in a pipe of uniform cross-sectional area A .



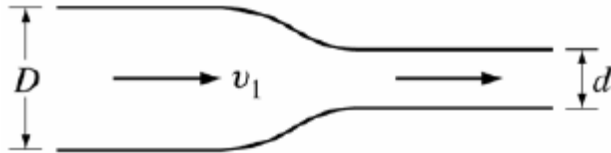
The pipe changes height from $y_1 = 2$ meters to $y_2 = 3$ meters. Since the areas are the same, we can say $v_1 = v_2$.

Which of the following is true?

- (A) $P_1 = P_2 + \rho g(y_2 - y_1)$
 - (B) $P_1 = P_2$
 - (C) $P_1 = 0$
 - (D) $P_2 = 0$
27. A vertical force of 30 N is applied uniformly to a flat button with a radius of 1 cm that is lying on a table. Which of the following is the best order of magnitude estimate for the pressure applied to the button?
- (A) 10^2 Pa
 - (B) 10^3 Pa
 - (C) 10^4 Pa
 - (D) 10^5 Pa

28. A ball that can float on water has mass 5.00 kg and volume $2.50 \times 10^{-2} \text{ m}^3$. What is the magnitude of the downward force that must be applied to the ball to hold it motionless and completely submerged in fresh water of density $1.00 \times 10^3 \text{ kg/m}^3$?
- (A) 20.0 N
(B) 25.0 N
(C) 30.0 N
(D) 200 N

29. Water flows through the pipe shown. At the larger end, the pipe has diameter D and the speed of the water is v_1 .



What is the speed of the water at the smaller end, where the pipe has diameter d ?

- (A) $\frac{d}{D} v_1$ (B) $\frac{D}{d} v_1$ (C) $\frac{d^2}{D^2} v_1$ (D) $\frac{D^2}{d^2} v_1$

AP Physics Free Response Practice – Fluids

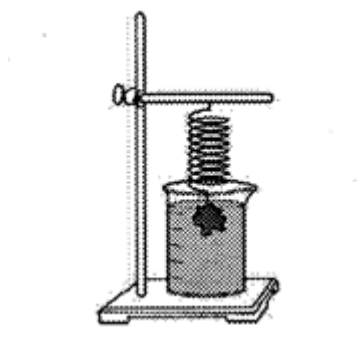
2002B6.

In the laboratory, you are given a cylindrical beaker containing a fluid and you are asked to determine the density ρ of the fluid. You are to use a spring of negligible mass and unknown spring constant k attached to a stand. An irregularly shaped object of known mass m and density D ($D \gg \rho$) hangs from the spring. You may also choose from among the following items to complete the task.

- A metric ruler
- A stopwatch
- String

(a) Explain how you could experimentally determine the spring constant k .

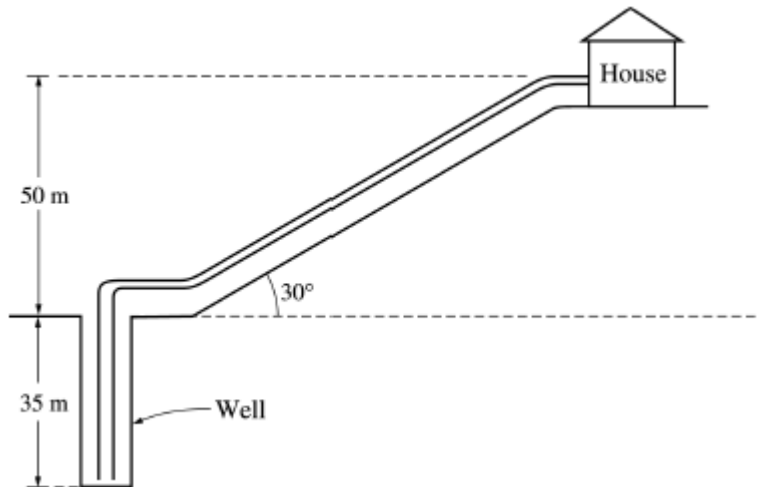
(b) The spring-object system is now arranged so that the object (but none of the spring) is immersed in the unknown fluid, as shown. Describe any changes that are observed in the spring-object system and explain why they occur.



(c) Explain how you could experimentally determine the density of the fluid.

(d) Show explicitly, using equations, how you will use your measurements to calculate the fluid density ρ . Start by identifying any symbols you use in your equations.

Symbol	Physical quantity



B2003B6.

A pump, submerged at the bottom of a well that is 35 m deep, is used to pump water uphill to a house that is 50 m above the top of the well, as shown above. The density of water is $1,000 \text{ kg/m}^3$. Neglect the effects of friction, turbulence, and viscosity.

- (a) Residents of the house use 0.35 m^3 of water per day. The day's pumping is completed in 2 hours during the day.
 - i. Calculate the minimum work required to pump the water used per day
 - ii. Calculate the minimum power rating of the pump.
- (b) In the well, the water flows at 0.50 m/s and the pipe has a diameter of 3.0 cm . At the house the diameter of the pipe is 1.25 cm .
 - i. Calculate the flow velocity at the house when a faucet in the house is open.
 - ii. Calculate the pressure at the well when the faucet in the house is open.

2003B6.

A diver descends from a salvage ship to the ocean floor at a depth of 35 m below the surface. The density of ocean water is $1.025 \times 10^3 \text{ kg/m}^3$.

- (a) Calculate the gauge pressure on the diver on the ocean floor.
- (b) Calculate the absolute pressure on the diver on the ocean floor.

The diver finds a rectangular aluminum plate having dimensions 1.0 m x 2.0 m x 0.03 m. A hoisting cable is lowered from the ship and the diver connects it to the plate. The density of aluminum is $2.7 \times 10^3 \text{ kg/m}^3$. Ignore the effects of viscosity.

- (c) Calculate the tension in the cable if it lifts the plate upward at a slow, constant velocity.
- (d) Will the tension in the hoisting cable increase, decrease, or remain the same if the plate accelerates upward at 0.05 m/s^2 ?

_____ increase _____ decrease _____ remain the same

Explain your reasoning.

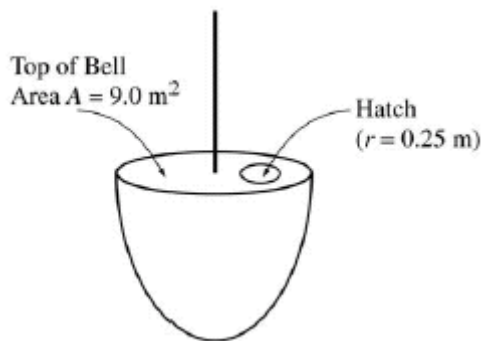
2004B2.

While exploring a sunken ocean liner, the principal researcher found the absolute pressure on the robot observation submarine at the level of the ship to be about 413 atmospheres. The inside of the submarine is kept at atmospheric pressure. The density of seawater is 1025 kg/m^3 .

- (a) Calculate the gauge pressure on the sunken ocean liner.
- (b) Calculate the depth of the sunken ocean liner.
- (c) Calculate the magnitude of the net force due to the fluid pressures only on a viewing port of the submarine at this depth if the viewing port has a surface area of 0.0100 m^2 .
- (d) What prevents the 'net force' found in part c from accelerating and moving the viewing port.

Suppose that the ocean liner came to rest at the surface of the ocean before it started to sink. Due to the resistance of the seawater, the sinking ocean liner then reached a terminal velocity of 10.0 m/s after falling for 30.0 s.

- (e) Determine the magnitude of the average acceleration of the ocean liner during this period of time.
- (f) Assuming the acceleration was constant, calculate the distance d below the surface at which the ocean liner reached this terminal velocity.
- (g) Calculate the time t it took the ocean liner to sink from the surface to the bottom of the ocean.



B2004B2.

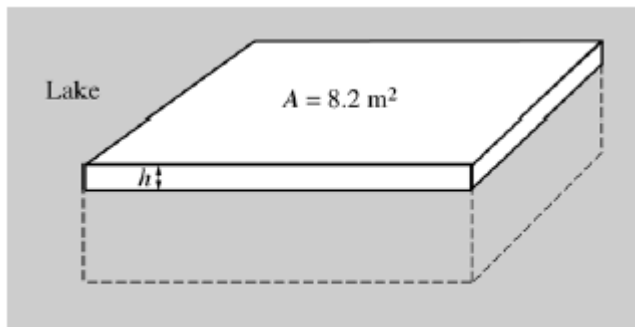
The experimental diving bell shown above is lowered from rest at the ocean's surface and reaches a maximum depth of 80 m. Initially it accelerates downward at a rate of 0.10 m/s^2 until it reaches a speed of 2.0 m/s , which then remains constant. During the descent, the pressure inside the bell remains constant at 1 atmosphere. The top of the bell has a cross-sectional area $A = 9.0 \text{ m}^2$. The density of seawater is 1025 kg/m^3 .

- Calculate the total time it takes the bell to reach the maximum depth of 80 m.
- Calculate the weight of the water on the top of the bell when it is at the maximum depth.
- Calculate the absolute pressure on the top of the bell at the maximum depth.

On the top of the bell there is a circular hatch of radius $r = 0.25 \text{ m}$.

- Calculate the minimum force necessary to lift open the hatch of the bell at the maximum depth.
- What could you do to reduce the force necessary to open the hatch at this depth? Justify your answer.

2005B5.



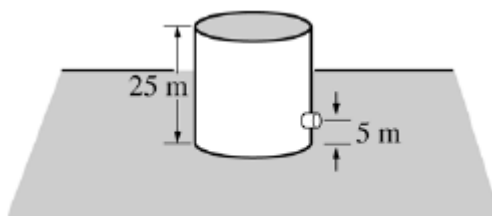
Note: Figure not drawn to scale.

A large rectangular raft (density 650 kg/m^3) is floating on a lake. The surface area of the top of the raft is 8.2 m^2 and its volume is 1.80 m^3 . The density of the lake water is 1000 kg/m^3 .

- Calculate the height h of the portion of the raft that is above the surrounding water.
- Calculate the magnitude of the buoyant force on the raft and state its direction.
- If the average mass of a person is 75 kg , calculate the maximum number of people that can be on the raft without the top of the raft sinking below the surface of the water. (Assume that the people are evenly distributed on the raft.)

B2005B5.

A large tank, 25 m in height and open at the top, is completely filled with saltwater (density 1025 kg/m^3). A small drain plug with a cross-sectional area of $4.0 \times 10^{-5} \text{ m}^2$ is located 5.0 m from the bottom of the tank.



The plug breaks loose from the tank, and water flows from the drain.

- Calculate the force exerted by the water on the plug before the plug breaks free.
- Calculate the speed of the water as it leaves the hole in the side of the tank.
- Calculate the volume flow rate of the water from the hole.

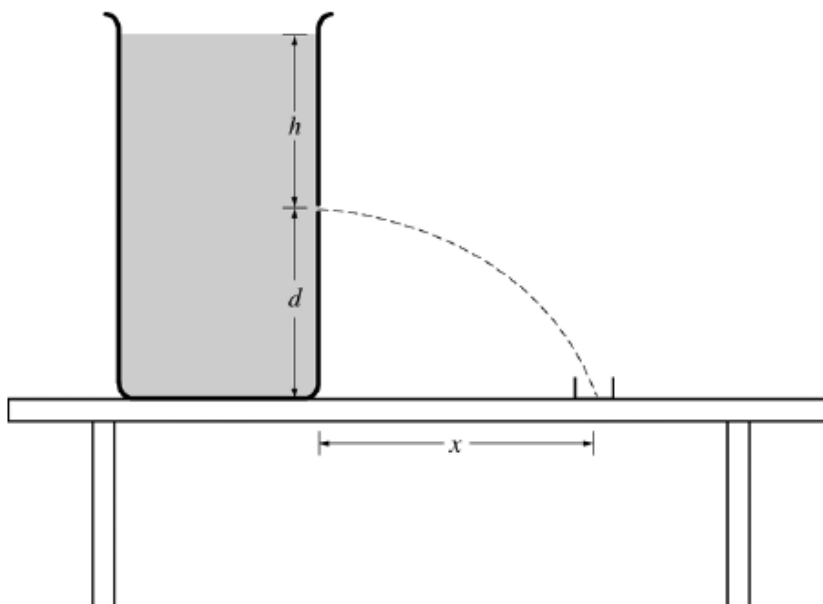
2007B4.

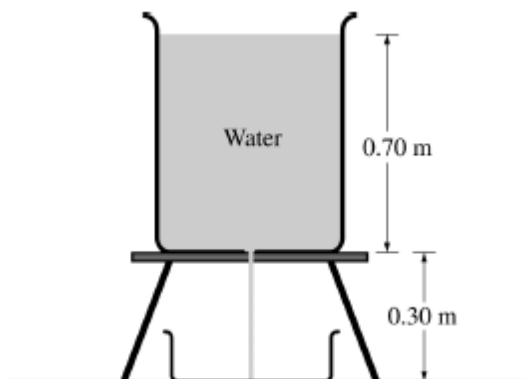
The large container shown in the cross section is filled with a liquid of density $1.1 \times 10^3 \text{ kg/m}^3$. A small hole of area $2.5 \times 10^{-6} \text{ m}^2$ is opened in the side of the container a distance h below the liquid surface, which allows a stream of liquid to flow through the hole and into a beaker placed to the right of the container. At the same time, liquid is also added to the container at an appropriate rate so that h remains constant. The amount of liquid collected in the beaker in 2.0 minutes is $7.2 \times 10^{-4} \text{ m}^3$.

- Calculate the volume rate of flow of liquid from the hole in m^3/s .
- Calculate the speed of the liquid as it exits from the hole.
- Calculate the height h of liquid needed above the hole to cause the speed you determined in part (b).
- Suppose that there is now less liquid in the container so that the height h is reduced to $h/2$. In relation to the collection beaker, where will the liquid hit the tabletop?

____ Left of the beaker ____ In the beaker ____ Right of the beaker

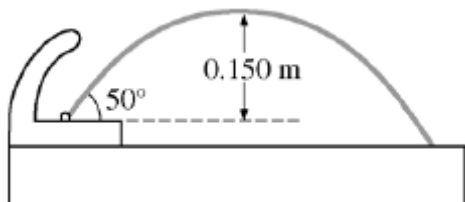
Justify your answer.



B2007B4.

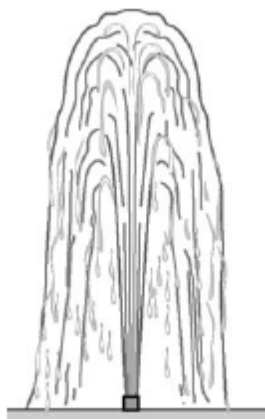
A cylindrical tank containing water of density 1000 kg/m^3 is filled to a height of 0.70 m and placed on a stand as shown in the cross section above. A hole of radius 0.0010 m in the bottom of the tank is opened. Water then flows through the hole and through an opening in the stand and is collected in a tray 0.30 m below the hole. At the same time, water is added to the tank at an appropriate rate so that the water level in the tank remains constant.

- (a) Calculate the speed at which the water flows out from the hole.
 - (b) Calculate the volume rate at which water flows out from the hole.
 - (c) Calculate the volume of water collected in the tray in $t = 2.0$ minutes.
 - (d) Calculate the time it takes for a given droplet of water to fall 0.25 m from the hole.
-

2008B4.

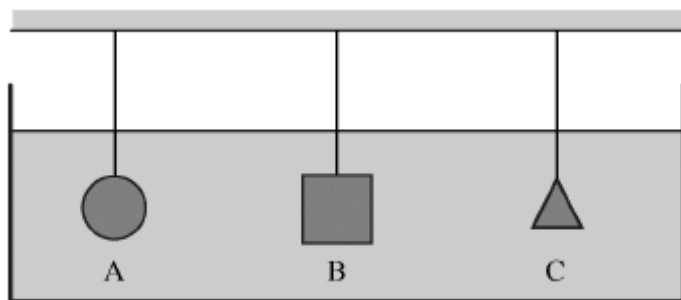
A drinking fountain projects water at an initial angle of 50° above the horizontal, and the water reaches a maximum height of 0.150 m above the point of exit. Assume air resistance is negligible.

- (a) Calculate the speed at which the water leaves the fountain.
- (b) The radius of the fountain's exit hole is $4.00 \times 10^{-3} \text{ m}$. Calculate the volume rate of flow of the water.
- (c) The fountain is fed by a pipe that at one point has a radius of $7.00 \times 10^{-3} \text{ m}$ and is 3.00 m below the fountain's opening. The density of water is $1.0 \times 10^3 \text{ kg/m}^3$. Calculate the gauge pressure in the feeder pipe at this point.

B2008B4.

A fountain with an opening of radius 0.015 m shoots a stream of water vertically from ground level at 6.0 m/s. The density of water is 1000 kg/m^3 .

- Calculate the volume rate of flow of water.
- The fountain is fed by a pipe that at one point has a radius of 0.025 m and is 2.5 m below the fountain's opening. Calculate the absolute pressure in the pipe at this point.
- The fountain owner wants to launch the water 4.0 m into the air with the same volume flow rate. A nozzle can be attached to change the size of the opening. Calculate the radius needed on this new nozzle.

2009B5.

Three objects of identical mass attached to strings are suspended in a large tank of liquid, as shown above.

- Must all three strings have the same tension?

___ Yes ___ No

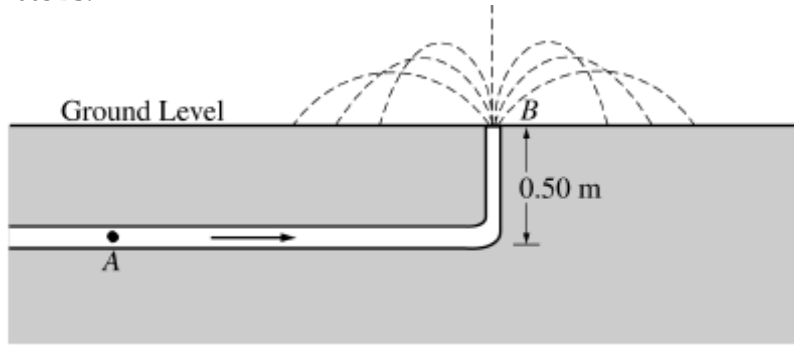
Justify your answer.

Object A has a volume of $1.0 \times 10^{-5} \text{ m}^3$ and a density of 1300 kg/m^3 . The tension in the string to which object A is attached is 0.0098 N.

- Calculate the buoyant force on object A.
- Calculate the density of the liquid.
- Some of the liquid is now drained from the tank until only half of the volume of object A is submerged. Would the tension in the string to which object A is attached increase, decrease, or remain the same?

___ Increase ___ Decrease ___ Remain the same

Justify your answer.

B2009B3.

An underground pipe carries water of density 1000 kg/m^3 to a fountain at ground level, as shown above. At point A, 0.50 m below ground level, the pipe has a cross-sectional area of $1.0 \times 10^{-4} \text{ m}^2$. At ground level, the pipe has a cross-sectional area of $0.50 \times 10^{-4} \text{ m}^2$. The water leaves the pipe at point B at a speed of 8.2 m/s .

- Calculate the speed of the water in the pipe at point A.
- Calculate the absolute water pressure in the pipe at point A.
- Calculate the maximum height above the ground that the water reaches upon leaving the pipe vertically at ground level, assuming air resistance is negligible.
- Calculate the horizontal distance from the pipe that is reached by water exiting the pipe at 60° from the level ground, assuming air resistance is negligible.

Supplemental Problems

SUP1. A block of wood has a mass of 12 kg and dimensions 0.5 m by 0.2 m by 0.2 m .

- Find the density ρ_o of the wooden block.
- If the block is placed in water ($\rho = 1000 \text{ kg/m}^3$) with the square sides parallel to the water surface, how far beneath the surface of the water is the bottom of the block?
- A weight is placed on the top of the block. The block sinks to a point that the top of the block is exactly even with the water surface. Find the mass of the added weight.

SUP2. A tapered horizontal pipe carries water from one building to another on the same level. The wider end has a cross-sectional area of 4 m^2 . The narrower end has a cross-sectional area of 2 m^2 . Water enters the wider end at a velocity of 10 m/sec .

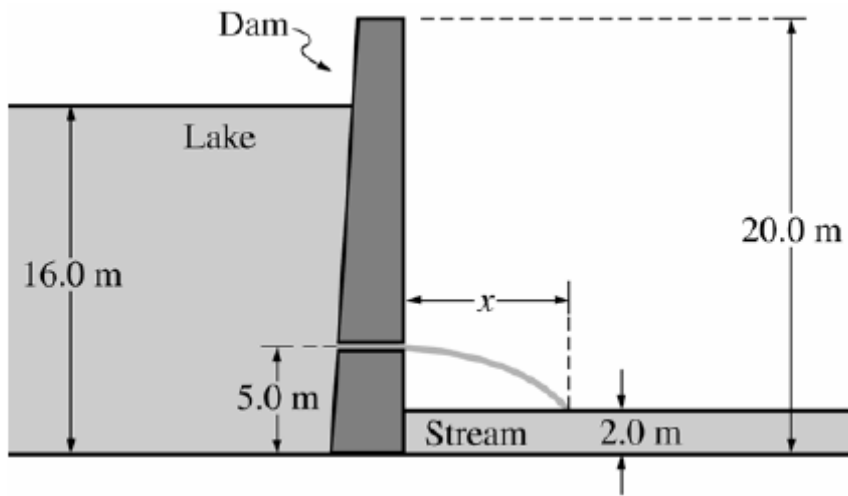
- What is the speed of the water at the narrow end of the pipe?
- The gauge pressure of the water at the wide end of the pipe is 2×10^5 pascals. Using Bernoulli's equation, find the gauge pressure at the narrow end of the pipe.

SUP3. A small airplane has wings with surface area 9 m^2 each. The speed of the air across the top of the wing is 50 m/sec , and across the bottom of the wing, 40 m/sec . Take the density of air to be 1.2 kg/m^3 .

- Find the difference in the pressure between the top and the bottom of the wing.
- Find the net lift upward on the plane.
 - If there is no other lift on the plane, what would be the mass of the plane? Assume the plane is not accelerating up or down.

SUP4. A block of wood floats in water, with $2/3$ of it submerged. The wood is then placed in oil, and $9/10$ of it is submerged. Find the density of the wood, and of the oil.

SUP5.



A 20 m high dam is used to create a large lake. The lake is filled to a depth of 16 m as shown above. The density of water is 1000 kg/m^3 .

(a) Calculate the absolute pressure at the bottom of the lake next to the dam.

A release valve is opened 5.0 m above the base of the dam, and water exits horizontally from the valve.

(b) Use Bernoulli's equation to calculate the initial speed of the water as it exits the valve.

(c) The stream below the surface of the dam is 2.0 m deep. Assuming that air resistance is negligible, calculate the horizontal distance x from the dam at which the water exiting the valve strikes the surface of the stream.

(d) Suppose that the atmospheric pressure in the vicinity of the dam increased. How would this affect the initial speed of the water as it exits the valve?

___ It would increase. ___ It would decrease. ___ It would remain the same.
Justify your answer.

ANSWERS - AP Physics Multiple Choice Practice – Fluids

Solution	Answer
1. FBD has F_t pointing down F_b pointing up and weight (mg) down. $F_{net} = 0$ $F_b - F_t - mg = 0$ The buoyant force is given by the weight of the displaced water. Since the water displaced volume is equal to the corks displaced volume and the water weight for the same volume would be 4 times heavier (based on the given cork weight = 25% water weight) compared to the cork, the buoyant force is equal to 4 x the cork weight = $4mg$. Using the force equation created initially. $F_t = F_b - mg = 4mg - mg = 3mg$	D
2. Using fluid continuity. $A_1 v_1 = A_2 v_2$ $\pi R^2 v_1 = \pi (2R)^2 v_2$ $v_1 = 4 v_2$	A
3. This is based on two principles. 1 – Bernoulli's principle says that when speed increases pressure drops. Second, continuity says more area means less speed based on $A_1 v_1 = A_2 v_2$ So the smallest area would have the largest speed and therefore most pressure drop.	B
4. Since A and B have the same mass and density, they have the same volume. C has the same volume as A and B since it's the same shape as B. So all three objects have the same volume. When submerged, they will all displace the same amount of water and therefore all have the same buoyant force acting on them. <i>Note: if the objects were floating instead of submerged then the heavier ones would have larger buoyant forces.</i>	D
5. Buoyant force is equal to weight of displaced fluid. Since the density is constant and the volume displaced is always the same, the buoyant force stays constant	B
6. The wood is floating and is only partially submerged. It does not displace a weight of water related to its entire volume. The iron however is totally submerged and does displace a weight of water equal to its entire volume. Since the iron displaces more water, it has a larger buoyant force acting on it.	B
7. For floating objects, the weight of the displaced fluid equals the weight of the object. For a more dense fluid, less of that fluid needs to be displaced to create a fluid weight equal to the weight of the object. Since the salt water is more dense, it will not need as much displaced.	B
8. More area $\rightarrow$ less speed $\rightarrow$ more pressure	B
9. Flow continuity. $A_1 v_1 = A_2 v_2$ $\pi (0.02)^2 (1) = \pi (0.01)^2 v_2$	D
10. Buoyant force is based on how much weight of water is displaced. Since all three are completely submerged they all displace the same amount of water so have equal buoyant forces.	C
11. For floating objects, the buoyant force equals the weight of the objects. Since each object has the same weight, they must have the same buoyant force to counteract that weight and make them float. <i>IF the equal mass objects sunk, then the one with the smaller density would have a larger volume and displace more water so have a larger buoyant force. But that is not the case here.</i>	D
12. Three forces act on the block, F_t down, mg down and F_b up. $F_{net} = 0$ $F_b - F_t - mg = 0$ $F_b - 3 - 5 = 0$ $F_b = 8 \text{ N} - \text{weight of displaced water} = \rho_{h20} V_{disp} g$ $8 = (1000) V (10) \rightarrow V = 0.0008 \text{ m}^3$	D
13. For floating objects $mg = F_b$ $\rho_{obj} V_{obj} g = \rho_{h20} V_{disp} g$ $\rho_{obj} (V)g = 1000 (0.6V) g$ $\rho_{obj} = 600$ In oil the same is true $\rho_{obj} V_{obj} g = \rho_{oil} V_{disp} g$ $(600)Vg = (800) x\%V g$ $x\% = 0.75$	D

14. Using fluid continuity. $A_1 v_1 = A_2 v_2$ $\pi R^2 v_1 = \pi (2R)^2 v_2$ $v_1 = 4 v_2$ A
15. Based on the continuity principle, less area means more speed and based on Bernoulli's principle, more speed means less pressure B
16. The weight of the object The weight of the mass is 4N. The scale reading apparent weight is 3N so there must be a 1N buoyant force acting to produce this result. B
17. . Since the pressure in a fluid is only dependent on the depth, they all have the same fluid pressure at the base. Since all of the bases have the same area and the same liquid pressure there, the force of the liquid given by $P=F/A$ would be the same for all containers. *Note: IF instead this question asked for the pressure of the container on the floor below it, the container with more total mass in it would create a greater pressure, but that is not the case here.* D
18. As the fluid flows into the smaller area constriction, its speed increases and therefore the pressure drops. Since the pressure in the constriction is less than that outside at the water surface, fluid is forced up into the lower tube. A,D
19. Both objects are more dense than water and will sink in the pool. Since both have the same volume, they will displace the same amount of water and will have the same buoyant forces. C
20. Again both samples sink. Also, both samples have the same mass but different densities. For the same mass, a smaller density must have a larger volume, and the larger volume displaces more water making a larger buoyant force. So the smaller density with the larger volume has a larger buoyant force. B
21. V of this ball is $\frac{4}{3} \pi r^3 = \frac{4}{3} \pi (0.4)^3 = 0.2681 \text{ m}^3$. For the ball to just sink, it is on the verge of floating, meaning the weight of the ball equals the buoyant force of the fully submerged ball.
 $mg = \rho_{\text{fl}} V_{\text{disp}} g$ $m (10) = 1400 (0.2681) (10)$ $m = 375 \text{ kg}$ B
22. Using fluid continuity. $A_1 v_1 = A_2 v_2$ $\pi (7R)^2 v_1 = \pi (R)^2 V$ $v_1 = V / 49$ A
23. The fluid flow is occurring in a situation similar to the diagram for question #27. Apply Bernoulli's equation. $P_1 + \rho g y_1 + \frac{1}{2} \rho v_1^2 = P_2 + \rho g y_2 + \frac{1}{2} \rho v_2^2$
 $P + 0 + \frac{1}{2} \rho v^2 = P_2 + \rho g y + \frac{1}{2} \rho (2v)^2$
 $P_2 = P + \frac{1}{2} \rho v^2 - \frac{1}{2} 4 \rho v^2 - \rho g y$ $= P - \frac{3}{2} \rho v^2 - \rho g y$ B
24. Definition of buoyant force C
25. The relevant equation is $P = P_o + \rho g h$
 Answer (a) is wrong, because at y_1 on both arms, the pressure is just the atmospheric pressure. The pressure in the right arm at y_3 is still just atmospheric, but on the left, it is atmospheric plus $\rho g (y_1 - y_3)$. That rules out (a). The pressure at the bottom of the tube is everywhere the same (Pascal's principle), which rules out (c), and at the same time, tells us (b) is right. At y_2 , we can say $P = P_{\text{bottom}} - \rho_{\text{Hg}} g y_2$ on both sides, so the pressure is equal. Answer (d) is wrong because at y_3 , the right arm is supporting only the atmosphere, while the left arm is supporting the atmosphere plus $\rho_{\text{H}_2\text{O}} g h$. Finally, (e) is silly because both arms at height y_1 are at atmospheric pressure. B,C
26. Apply Bernoulli's equation. $P_1 + \rho g y_1 + \frac{1}{2} \rho v_1^2 = P_2 + \rho g y_2 + \frac{1}{2} \rho v_2^2$
 $P_1 = P_2 + \rho g (y_2 - y_1)$ A

27. $P = F / A = 30 / \pi r^2 \dots$ use 3 for π since its an estimate ... $30 / (3*(.01)^2) = 100000 \text{ Pa}$ D
28. For the object to be completely submerged there would be three forces acting. F_b up, mg down and F_{push} down. $F_b = F_{\text{push}} + mg$ $F_{\text{push}} = F_b - mg$ D
 $F_{\text{push}} = \rho_{\text{h20}} V_{\text{disp}} g - mg$
 $= (1000)(2.5 \times 10^{-2})(10) - (5)(10) = 200 \text{ N}$
29. Using fluid continuity. $A_1 v_1 = A_2 v_2$ $\pi(D/2)^2 v_1 = \pi(d/2)^2 v_2$ solve for v_2 D

AP Physics Free Response Practice – Fluids – ANSWERS

2002B6.

a) Example 1: Measure the unstretched length of the spring. Hang it with the object at rest and measure the stretched length. Call the difference in these lengths Δx . Equating the weight of the object and the force exerted by the extended spring gives $mg = k\Delta x$ from which k can be determined.

Example 2: Set the hanging mass into oscillation. Determine the period T by timing n oscillations and dividing that time by n . The equation $T = 2\pi\sqrt{m/k}$ can then be used to find k .

b) The spring is stretched less when the object is at rest in the fluid. The fluid exerts an upward buoyant force on the object. Since the net force on the object is still zero, the spring does not need to exert as much force as before and thus stretches less.

c&d)

1) Measure the length of the spring when the object is immersed in the liquid, and subtract the unstretched length to determine the amount the spring is stretched. This will allow calculation of the force exerted by the spring on the object.

2) The volume of fluid displaced is equal to the volume of the object, which can be determined from the given mass and density of the object.

3) The buoyant force on the object is equal to the difference of the object's weight and the force exerted by the spring.

4) The buoyant force also equals the weight of the displaced fluid, which equals the product of the fluid density, displaced volume, and g .

<u>Symbol</u>	<u>Quantity</u>
ρ	fluid density
V	object volume = displaced water volume
g	acceleration of gravity
m	mass of object
x	spring stretch in air
x_w	spring stretch in water

First solving for k in air. $mg = kx$

Then in the fluid. $F_{sp} = kx_w$

$$F_{net} = 0 \quad F_b = mg - F_{sp} \quad \rho Vg = mg - kx_w \quad \rho Vg = mg - (mg/x) x_w \quad \text{solve for } \rho$$

B2003B6.

- a) i) The total mass of water moved can be found with the density and volume $m = \rho V = (1000)(0.35) = 350$ kg of water. This water is moved a distance 85 m so the work done to move it is $W = Fd = (350)(9.8)(85) = 291,500$ J.

ii) The force needed to move the water = the weight of the water (mg).

Using. $P = Fd / t = (350)(9.8)(85) / (2\text{hrs} * 3600 \text{ s/hr}) = 40.5$ W

- b) i) Using fluid continuity. $A_1 v_1 = A_2 v_2$ $\pi(.03/2)^2(0.5) = \pi(0.0125/2)^2 v_2$ $v_2 = 2.88$ m/s

ii) Apply Bernoulli's equation. $P_1 + \rho g y_1 + \frac{1}{2} \rho v_1^2 = P_2 + \rho g y_2 + \frac{1}{2} \rho v_2^2$
 $y_1 = 0$ $P_2 = \text{atmospheric (open faucet)}$

$$P_1 + 0 + \frac{1}{2} (1000) (0.5)^2 = 1.01 \times 10^5 + (1000)(9.8)(85) + \frac{1}{2} (1000)(2.88)^2 \quad P_1 = 938000 \text{ Pa}$$

2003B6.

a) $P = \rho g h = (1025)(9.8)(35) = 351,600$ Pa

b) $P_{\text{abs}} = P_o + \rho g h = 1.01 \times 10^5 + 343,000 = 452,600$ Pa

- c) The FBD has three forces acting on it. The upwards lifting force, the upwards buoyant force and the downwards weight, mg. Constant velocity $\rightarrow F_{\text{net}} = 0$

$$F_t + F_b - mg = 0 \quad F_t = mg - F_b \quad F_t = (\rho_{\text{obj}} V_{\text{obj}})g - (\rho_{\text{h20}} V_{\text{disp}})g \quad V_{\text{disp}} = V_{\text{obj}} \text{ call it } V$$

$$\rightarrow F_t = Vg (\rho_{\text{al}} - \rho_{\text{h20}}) = (1 \times 2 \times 0.03)(9.8)(2700 - 1025) = 985 \text{ N}$$

- d) $F_t + F_b - mg = ma$ $F_t = mg - F_b + ma$. Comparing this tension equation to the one in part c you see that the tension will increase since the quantity "ma" is being added here
-

2004B2.

a) $P_{\text{abs}} = P_o + P_{\text{gauge}} \quad 413 \text{ atm} = 1 \text{ atm} + P_{\text{gauge}} \quad P_{\text{gauge}} = 412 \text{ atm}$

b) $P_{\text{gauge}} = \rho gh \quad 412(1.01 \times 10^5) = 1024(9.8)(h) \quad h = 4140 \text{ m}$

- c) The fluid pressures acting are the outside water pressure (which includes the atmosphere at the surface acting down on it) and, the inside air pressure which is atmospheric. Since the atmospheric pressure acts both inside and is also included in the water pressure, the net force due to fluid pressure can be found by using the water's gauge pressure since the air pressures effectively cancel each other out.

$$P_{\text{gauge}} = F/A \quad 412(1.01 \times 10^5) = F/0.01 \quad F = 416,000 \text{ N}$$

- d) The force from c is not the true net force. The actual net force is zero as the window is at rest. This force from c is due to fluid pressures and is resisted by normal forces acting on the edges of the window where it is connected to the submarine.

e) $v_f = v_i + at \quad 10 = 0 + a(30) \quad a = 0.33 \text{ m/s}^2$

f) $d = v_i t + \frac{1}{2} at^2 \quad d = 0 + \frac{1}{2}(0.33)(30)^2 \quad d = 150 \text{ m}$

- g) The total depth is 4140 m. There are two parts to the trip, the first 150 m covered while accelerating and the second (3990m) covered while moving at constant speed. The parts must be calculated separately. Part one, during acceleration, was already given as taking 30 second. The second part at a constant speed can simply be found using $\bar{v} = d/t$, $10 = 3990/t$, $t_2 = 399$ seconds. So the total time of travel was 429 seconds.

B2004B2.

- a) The descent occurs at two different accelerations and must be analyzed in the two sections.

Section 1 starts from rest and accelerates, find the time in that part

$$v_{1f} = v_{1i} + a_1 t_1 \quad 2 = 0 + 0.10 t \quad t_1 = 20 \text{ seconds.}$$

$$d_1 = v_{1i} t + \frac{1}{2} a_1 t_1^2 \quad d_1 = \frac{1}{2}(0.10)(20)^2 \quad d_1 = 20 \text{ m}$$

Section 2 occurs at a constant speed equal to the final speed in section 1 and will occur over the remaining distance $d_2 = 60\text{m}$.

$$\bar{v}_2 = d_2 / t_2 \quad 2 = 60 / t_2 \quad t_2 = 30 \text{ seconds} \quad t_{\text{total}} = t_1 + t_2 = 50 \text{ seconds}$$

- b) Weight of water above the bell is a cylindrical column with a height of $h=80 \text{ m}$ and area of $A=9 \text{ m}^2$. This gives us the volume of the water above the bell given by $V = Ah = 720 \text{ m}^3$.

$$\text{The weight of this column} = m_{h20} g = (\rho_{h20} V) g = (1025)(720)(9.8) = 7.2 \times 10^5 \text{ N}$$

c) $P_{\text{abs}} = P_o + \rho gh = 1.01 \times 10^5 + (1025)(9.8)(80) = 9 \times 10^5 \text{ Pa}$

- d) Since there is air pressure inside the bell, and the absolute pressure on the outside also includes the air pressure, these two pressures essentially cancel each other out and we only need to push against the water pressure alone so we should use the gauge pressure to find the needed force.

$$P_{\text{abs}} = P_o + P_{\text{gauge}} \quad 9 \times 10^5 = 1.01 \times 10^5 + P_{\text{gauge}} \quad P_{\text{gauge}} = 8 \times 10^5 \text{ Pa.}$$

$$F = PA = (8 \times 10^5)(\pi(0.25)^2) = 1.58 \times 10^5 \text{ N}$$

- e) To reduce the pushing force needed, you could increase the pressure inside the bell to create a smaller pressure difference between inside and outside. Or, by making the area of the hatch smaller the pushing force would be less. Or, you could use a lever inside that uses torque to provide mechanical advantage to amplify an applied force to one side of the lever. This would make the force pushing the hatch open the same but the required pushing force of a person less.

2005B5.

- a) We are given the volume of the raft and the surface area as well. Use this to first find the total height of the raft h_t
 $V = Ah_t$ $1.8 = 8.2 h_t$ $h_t = 0.22 \text{ m}$

Since the raft is floating, the weight of the raft must equal the weight of the displaced fluid. We will define " h_s " as being the portion of the height of the raft below the water so that the displaced volume is given by $V = Ah_s$

$$m_{\text{raft}} g = \rho_{\text{h20}} V_{\text{disp}} g$$

$$\rho_{\text{raft}} V_{\text{raft}} g = \rho_{\text{h20}} V_{\text{disp}} g \quad \rho_{\text{raft}} V_{\text{raft}} = \rho_{\text{h20}} (Ah_s) \quad (650)(1.8) = (1000)(8.2)h_s \quad h_s = 0.143 \text{ m}$$

$$h = h_t - h_s = 0.22 - 0.143 = 0.077 \text{ m (the visible portion of the raft)}$$

- b) F_B equals weight of displaced water $= \rho_{\text{h20}} V_{\text{disp}} g = \rho_{\text{h20}} (Ah_s) g = (1000)(8.2)(0.143)(9.8) = 11500 \text{ N}$ directed $\uparrow$

- c) Determine the extra buoyant force that will come from submerging the exposed raft volume $V_{\text{exp}} = Ah$

$$F_{\text{b(extra)}} = \rho_{\text{h20}} V_{\text{disp}} g = \rho_{\text{h20}} (Ah) g = (1000)(8.2)(0.077)(9.8) = 6187.7 \text{ N}$$

$$1 \text{ persons weight} = mg = 735 \text{ N. Total weight allowed / person weight} = 6187.7 / 735 = 8.41$$

So, an extra 8 people could come on without submerging the raft. You could also chop some arms or legs off and throw them on there also until you get up to the extra 0.41 of a person limit.

B2005B5.

- a) The force on the plug from the water inherently includes the atmosphere above it, so we use the absolute pressure.

$$P_{\text{abs}} = P_o + \rho gh = 1.01 \times 10^5 + (1025)(9.8)(20\text{m}) = 3 \times 10^5 \text{ Pa}$$

$$\text{The force is then found with } P = F/A \quad 3 \times 10^5 = F / (4 \times 10^{-5}) \rightarrow F = 12 \text{ N}$$

Note: This calculation of pressure (ρgh) only works since the fluid is at rest (static). For moving fluids, only Bernoulli's equation (or F/A in rare cases) can be applied for determining pressures.

- b) Though many of you may know the Torricelli theorem shortcut to this problem, when the AP exam graded this question, simply stating that equation and plugging in lost points. To be safe you should always start with Bernoulli's equation in its full form, cancel out terms that don't exist or are assumed zero, and solve from there.

$$P_1 + \rho gy_1 + \frac{1}{2} \rho v_1^2 = P_2 + \rho gy_2 + \frac{1}{2} \rho v_2^2$$

P_1 and P_2 are both open to atmosphere so are at P_o and cancel. Tank is large so v_1 is assumed small enough to be 0, y_2 is set as zero height.

$$\rho gy_1 = \frac{1}{2} \rho v_2^2 \quad v_2 = \sqrt{2gy_1} \text{ (as expected from Torricelli) } \dots v_2 = \sqrt{2(9.8)(20)} = 19.8 \text{ m/s}$$

- c) Volume flow rate $= Q = Av = (4 \times 10^{-5})(19.8) = 7.92 \times 10^{-4} \text{ m}^3/\text{s}$

2007B4.

- a) Volume flow rate $= Q = V/t = 7.2 \times 10^{-4} / (2\text{min} * 60 \text{ sec/min}) = 6 \times 10^{-6} \text{ m}^3/\text{s}$

- b) Your first thought is probably Bernoulli, but there are too many unknowns so this does not work. We can use the volume flow rate above to find the velocity.

$$Q = Av \quad 6 \times 10^{-6} = (2.5 \times 10^{-6}) v \quad v = 2.4 \text{ m/s}$$

- c) Use Bernoulli, same derivation as in the problem above (B2005B5) $\dots v_2 = \sqrt{2gh} \quad (2.4) = \sqrt{2(9.8)h} \quad h = 0.29 \text{ m}$

- d) Left of beaker. Based on the formula derived above, the exit velocity is dependent on the height and with less horizontal exit velocity the range will be less ($d_x = v_x t$). This makes sense because less height would result in less pressure and decrease the speed the fluid is ejected at, thus lessening the range.

B2007B4.

a) Use Bernoulli, same derivation as problem B2005B5 ... $v_2 = \sqrt{2gh}$... $v_2 = \sqrt{2(9.8)(0.7)} \dots v_2 = 3.7 \text{ m/s}$

b) Volume flow rate $Q = Av = \pi(0.001)^2(3.7) = 1.16 \times 10^{-5} \text{ m}^3/\text{s}$

c) $Q = V / t$ $1.16 \times 10^{-5} = V / (2 \text{ min} * 60 \text{ s/min})$ $V = 0.0014 \text{ m}^3$

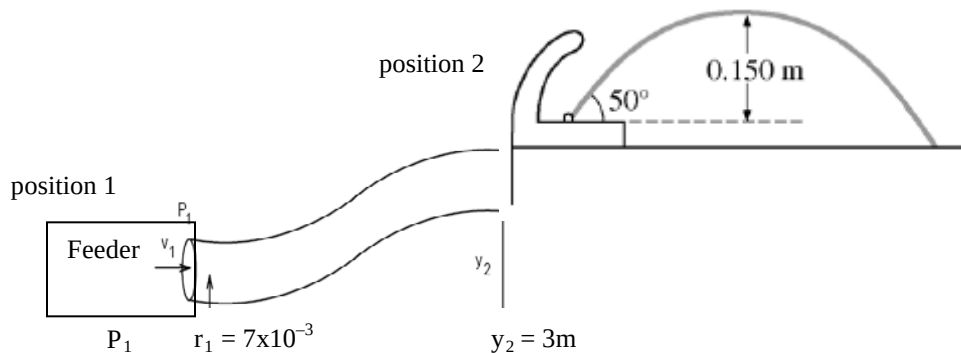
d) Free fall. $d = v_i t + \frac{1}{2} g t^2$ $-0.25 = (-3.7 t) + \frac{1}{2}(-9.8) t^2$ solve quadratic $t = 0.062 \text{ s}$
 Alternatively, first determine v_f at the 0.25 m location then use $v_f = v_i + at$ to solve for t .

2008B4.

a) Using projectile methods. $v_{fy}^2 = v_{iy}^2 + 2ad_y$ $0 = (v_i \sin 50^\circ)^2 + 2(-9.8)(0.15)$ $v_i = 2.24 \text{ m/s}$

b) Volume flow rate $= Q = Av = \pi(4 \times 10^{-3})^2 (2.24) = 1.13 \times 10^{-4} \text{ m}^3/\text{s}$

c) If you don't understand the wording, here is what the problem is saying



First we need to find the velocity of the water at the feeder using continuity

$$A_1 v_1 = Q_2 \quad \pi(7 \times 10^{-3})^2 (v_1) = 1.13 \times 10^{-4} \quad v_1 = 0.73 \text{ m/s}$$

Bernoulli. Position 2 is the fountain spigot which is open so at atmospheric pressure. $y_1 = 0$ no height.

$$P_1 + \rho g y_1 + \frac{1}{2} \rho v_1^2 = P_2 + \rho g y_2 + \frac{1}{2} \rho v_2^2$$

$$P_1 + 0 + \frac{1}{2} (1000)(0.73)^2 = (1.01 \times 10^5) + (1000)(9.8)(3 \text{ m}) + \frac{1}{2} (1000)(2.24)^2$$

$$P_1 = 1.32 \times 10^5 \text{ Pa which is the absolute pressure of the feeder.}$$

To find the gauge pressure of the feeder. $P_{\text{abs}} = P_{\text{gauge}} + P_o$ $1.32 \times 10^5 = P_{\text{gauge}} + 1.01 \times 10^5$

$$P_{\text{gauge}} = 31600 \text{ Pa.}$$

Note: This gauge pressure could be determined directly in Bernoulli's equation by realizing that P_1 includes atmospheric pressure as part of its total value and that P_2 was equal to atmospheric pressure, so by elimination of the term P_2 , P_1 becomes the gauge pressure. This should be stated in the solution if it is the chosen solution method.

B2008B4

a) Volume flow rate = $Q = Av = \pi(0.015)^2 (6) = 0.0042 \text{ m}^3/\text{s}$

b) First we need to find the velocity of the water in the pipe below using continuity
 $A_1 v_1 = Q_2 \quad \pi(0.025)^2(v_1) = 0.0042 \quad v_1 = 2.16 \text{ m/s}$

Bernoulli. Position 2 is the fountain spigot which is open so at atmospheric pressure. $y_1=0$ no height.

$$P_1 + \rho g y_1 + \frac{1}{2} \rho v_1^2 = P_2 + \rho g y_2 + \frac{1}{2} \rho v_2^2$$

$$P_1 + 0 + \frac{1}{2} (1000)(2.16)^2 = (1.01 \times 10^5) + (1000)(9.8)(2.5\text{m}) + \frac{1}{2} (1000)(6)^2 \quad P_1 = 141000 \text{ Pa}$$

c) Determine the launch speed needed to reach 4m.

Free fall of a water droplet. $v_f^2 = v_i^2 + 2gd \quad (0) = v_i^2 + 2(-9.8)(4) \quad v_i = 8.85 \text{ m/s}$

Use flow rate to find new area needed. $Q = Av \quad (0.0042) = A (8.85) \quad A_{\text{new}} = 4.75 \times 10^{-4} \text{ m}^2$

Find new radius $A_{\text{new}} = \pi r_{\text{new}}^2 \quad 4.75 \times 10^{-4} \text{ m}^2 = \pi r_{\text{new}}^2 \quad r_{\text{new}} = 0.0122 \text{ m}$

2009B5.

a) There are three forces acting on the masses in each case. Tension up, buoyant force up, weight down. Since they are at rest we have. $F_{\text{net}} = 0 \quad F_t + F_b = mg \quad F_t = mg - F_b$ so the largest F_b makes the largest F_t

We are to assume the diagram is to scale and that clearly the volumes of the three containers are different. The one with the largest volume displaces the largest amount and weight of water and will have the largest buoyant force acting on it. So since they all displace different volumes (and weights) of water they all have different buoyant forces, and based on the equation shown above will have different tensions.

b) The mass of the object is given by $m = \rho_{\text{obj}} V_{\text{obj}}$.

Using the equation from part a,

$$F_t + F_b = mg, \quad F_t + F_b = (\rho_{\text{obj}} V_{\text{obj}}) g \quad (0.0098) + F_b = (1300)(1 \times 10^{-5})(9.8) \quad F_b = 0.1176 \text{ N}$$

c) The buoyant force is by definition equal to the weight of the displaced fluid.

$$F_b = (\rho_{\text{fluid}} V_{\text{disp.}}) g \quad 0.1176 = \rho_{\text{fluid}} (1 \times 10^{-5})(9.8) \quad \rho_{\text{fluid}} = 1200 \text{ kg/m}^3$$

d) With only half of the volume submerged, $\frac{1}{2}$ as much water will be displaced and the buoyant force will be half the size. Based on the formula from part A, less buoyant force will make a larger tension. This also makes sense conceptually. Objects have large apparent weights in air than water so having some of it in the air will increase its apparent weight.

B2009B3.

- a) Using fluid continuity. $A_1 v_1 = A_2 v_2$ $(1 \times 10^{-4})(v_1) = (0.5 \times 10^{-4})(8.2)$ $v_a = 4.1 \text{ m/s}$
- b) Bernoulli. Position B is the fountain spigot which is open so at atmospheric pressure. $y_1 = 0$ no height.
 $P_1 + \rho g y_1 + \frac{1}{2} \rho v_1^2 = P_2 + \rho g y_2 + \frac{1}{2} \rho v_2^2$
 $P_1 + 0 + \frac{1}{2} (1000)(4.1)^2 = (1.01 \times 10^5) + (1000)(9.8)(0.5\text{m}) + \frac{1}{2} (1000)(8.2)^2$ $P_1 = 1.3 \times 10^5 \text{ Pa}$
- c) Free fall of a water droplet. $v_f^2 = v_i^2 + 2gd$ $(0)^2 = (8.2)^2 + 2(-9.8)(d)$ $d = 3.43\text{m}$
- d) Projectile method, in y direction.
 $d_y = v_{iy}t + \frac{1}{2}gt^2$ $d_y = (v_i \sin \theta)t + \frac{1}{2}gt^2$ $0 = (8.2 \sin 60)t + \frac{1}{2}(-9.8)t^2$ $t = 1.45 \text{ sec}$
- X direction. $d_x = v_x t$ $d_x = (v_i \cos \theta)t$ $d_x = (8.2 \cos 60)(1.45)$ $d_x = 5.95 \text{ m}$

SUP1.

- a) $\rho = m/V$ $= 12 / (0.5 \times 0.2 \times 0.2)$ $= 600 \text{ kg / m}^3$
- b) The block will float based on its density. For floating, block weight = buoyant force.
 $m_{\text{obj}}g = \rho_{\text{h20}} V_{\text{disp}} g$ $m_{\text{obj}} = \rho_{\text{h20}} A_{\text{square}}(h_{\text{submerged}})$ $12 = 1000(0.2 \times 0.2)h_{\text{sub}}$ $h_{\text{sub}} = 0.3 \text{ m}$
- c) The extra weight added should equal the extra buoyant force created by submerging the remaining 0.2 m of height.
 $F_{\text{b(extra)}} = \rho_{\text{h20}} V_{\text{disp}} g = (1000)(0.2 \times 0.2 \times 0.2)(9.8) = 78.4 \text{ N}$ $78.4 / 9.8 = 8\text{kg of extra mass.}$

SUP2.

- a) Using fluid continuity. $A_1 v_1 = A_2 v_2$ $(4)(10) = (2)(v_2)$ $v_2 = 20 \text{ m/s}$
- b) Bernoulli. $\rho g y_1$ terms cancel out since the pipe stays on the same level.
 $P_1 + \rho g y_1 + \frac{1}{2} \rho v_1^2 = P_2 + \rho g y_2 + \frac{1}{2} \rho v_2^2$
 $2 \times 10^5 + 0 + \frac{1}{2} (1000)(10)^2 = P_2 + 0 + \frac{1}{2} (1000)(20)^2$ $P_2 = 50000 \text{ Pa.}$
- Since P_1 was the gauge pressure and did not include P_o , P_2 will also come out as the gauge pressure.*

SUP3.

- a) Bernoulli. $\rho g y_1$ terms cancel out since the height difference is negligible.
 $P_1 + \rho g y_1 + \frac{1}{2} \rho v_1^2 = P_2 + \rho g y_2 + \frac{1}{2} \rho v_2^2$... rearrange equation so we can find $P_2 - P_1$ which is the ΔP
 $P_2 - P_1 = \frac{1}{2} \rho v_1^2 - \frac{1}{2} \rho v_2^2$ $\Delta P = \frac{1}{2} (1.2) (50^2 - 40^2) = 540 \text{ Pa}$
- b) i) $\Delta P = F_{\text{lift}} / A$ $540 = F_{\text{lift}} / (9 \times 2 \text{ wings})$ $F_{\text{lift}} = 9720 \text{ N}$
- ii) $F_{\text{net}} = 0$ $F_{\text{lift}} = mg$ $9720 = m(9.8)$ $m = 992 \text{ kg.}$

SUP4.

This problem involves floating objects, so weight of object = buoyant force $m_{\text{obj}} g = \rho_{\text{fluid}} V_{\text{disp}} g$

In general ... $m_{\text{obj}} = \rho_{\text{obj}} V_{\text{obj}}$

Giving ... $\rho_{\text{obj}} V_{\text{obj}} g = \rho_{\text{fluid}} V_{\text{disp}} g \quad \dots \quad \rho_{\text{obj}} V_{\text{obj}} = \rho_{\text{fluid}} V_{\text{disp}}$

Water

$$\rho_{\text{obj}} V_{\text{obj}} = \rho_{\text{fluid}} V_{\text{disp}}$$

$$\rho_w V = (1000)(2/3 V)$$

$$\rho_w = 666.67 \text{ kg / m}^3$$

Oil

$$\rho_{\text{obj}} V_{\text{obj}} = \rho_{\text{fluid}} V_{\text{disp}}$$

$$(666.67)V = \rho_{\text{oil}} (9/10 V)$$

$$\rho_{\text{oil}} = 740.74 \text{ kg / m}^3$$

SUP5.

a) $P_{\text{abs}} = P_{\text{gauge}} + P_o \quad P_{\text{abs}} = \rho gh + 1.01 \times 10^5 \quad P_{\text{abs}} = (1000)(9.8)(16) + 1.01 \times 10^5 = \quad 260000 \text{ Pa}$

b) Use Bernoulli, same derivation as problem B2005B5 ... $v_2 = \sqrt{2gh} \dots v_2 = \sqrt{2(9.8)(11)} \dots \quad v_2 = 14.7 \text{ m/s}$

c) Using projectile methods. $d_y = v_{iy}t + \frac{1}{2}at^2 \quad -3 = 0 + \frac{1}{2}(-9.8)t^2 \quad t = 0.78 \text{ sec}$

$$d_x = v_x t = (14.7)(0.78) = 11.5 \text{ m}$$

d) An increase in atmospheric pressure around the damn increases both P_1 and P_2 equally so there is no net effect on these terms in Bernoulli's equation, which means the exit velocity would be the same.